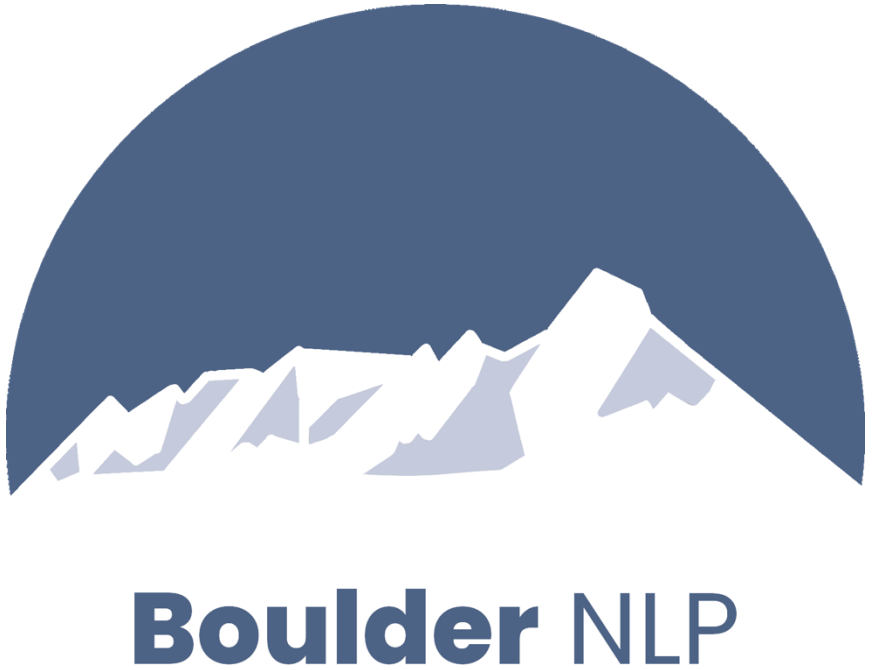
Natural Language Processing



Jim Martin -- CSCI 5832

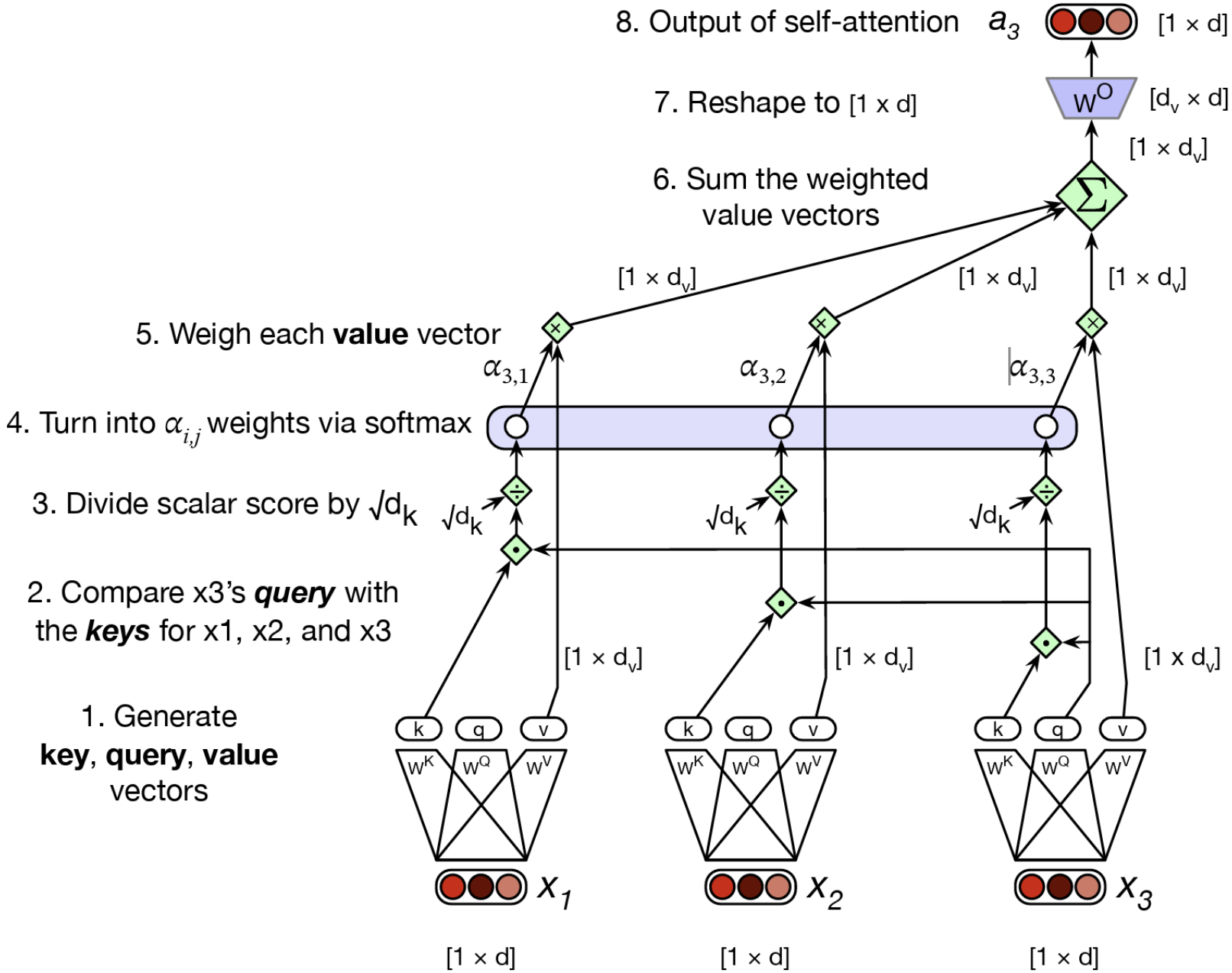
Lecture 17

Readings

- Embeddings
  - Chapter 5
- NNs
  - Chapter 6
- Transformers
  - Chapter 7

- Quiz 2: 10/27

Attention

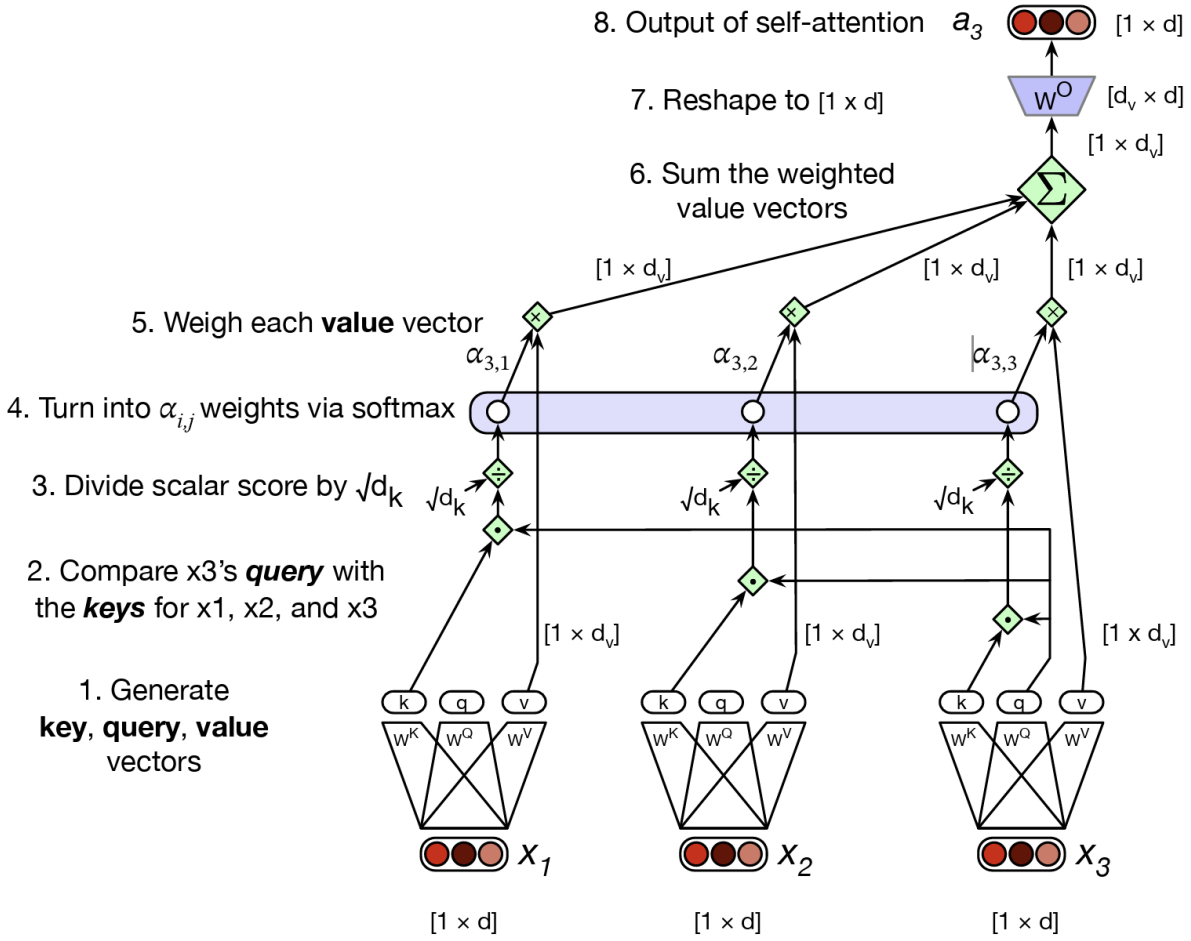

8. Output of self-attention a_3
[1 × d]
7. Reshape to [1 x d]
W^O
[d_v × d]
[1 × d_v]
6. Sum the weighted value vectors
[1 × d_v]
[1 × d_v]
[1 × d_v]
5. Weigh each **value** vector
$\alpha_{3,1}$
$\alpha_{3,2}$
$\alpha_{3,3}$
4. Turn into $\alpha_{i,j}$ weights via softmax
3. Divide scalar score by $\sqrt{d_k}$
$\sqrt{d_k}$
$\sqrt{d_k}$
$\sqrt{d_k}$
2. Compare x3's ***query*** with the ***keys*** for x1, x2, and x3
[1 × d_v]
[1 × d_v]
[1 x d_v]
1. Generate **key**, **query**, **value** vectors
k
q
v
W^K
W^Q
W^V
x_1
x_2
x_3
[1 × d]
[1 × d]
[1 × d]

Attention

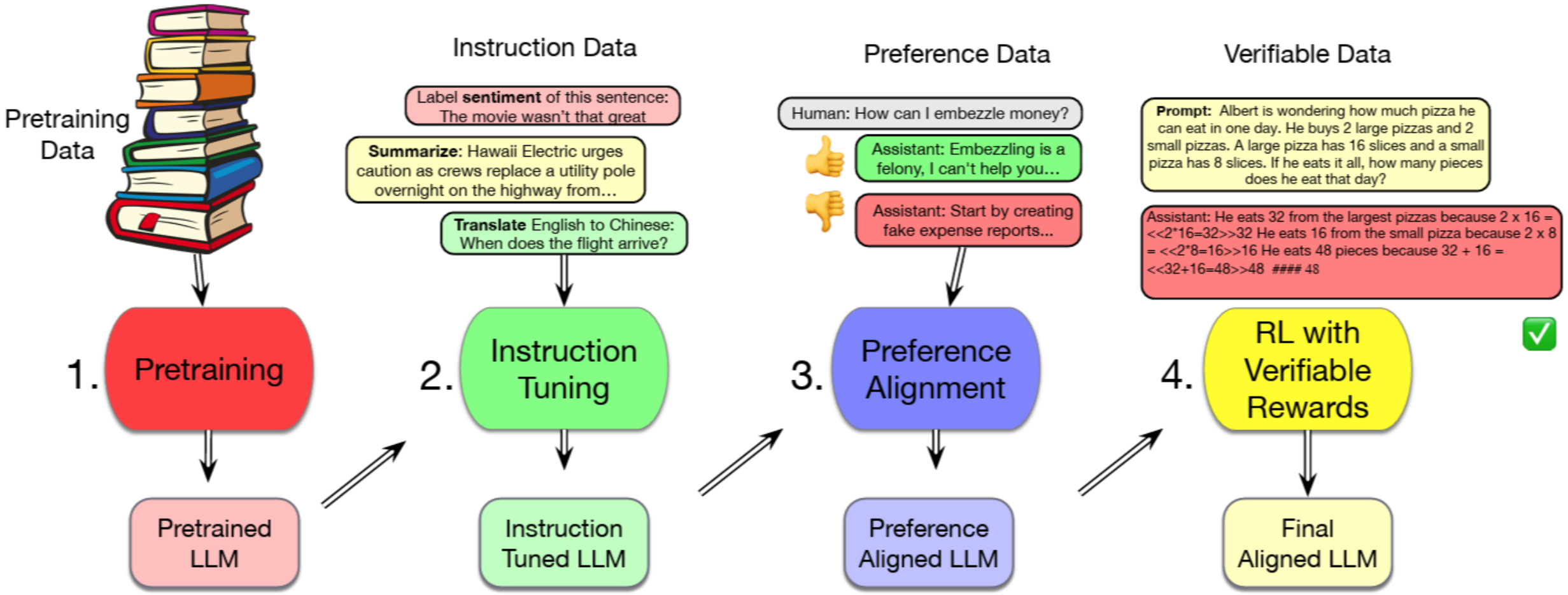
Model dimension = d = 1000

d_v, d_d_q, d_k = 125

How many weights are there to be learned for this attention calculation?



Pretraining/Post-training Pipeline



Instruction Data: Extractive Q/A

- **Definition:** This task involves creating answers to complex questions, from a given passage. Answering these questions, typically involve understanding multiple sentences. Make sure that your answer has the same type as the "answer type" mentioned in input. The provided "answer type" can be of any of the following types: "span", "date", "number". A "span" answer is a continuous phrase taken directly from the passage or question. You can directly copy-paste the text from the passage or the question for span type answers. If you find multiple spans, please add them all as a comma separated list. Please restrict each span to five words. A "number" type answer can include a digit specifying an actual value. For "date" type answers, use DD MM YYYY format e.g. 11 Jan 1992. If full date is not available in the passage you can write partial date such as 1992 or Jan 1992.
- **Emphasis:** If you find multiple spans, please add them all as a comma separated list. Please restrict each span to five words.
- **Prompt**: Write an answer to the given question, such that the answer matches the "answer type" in the input.
  Passage: { passage}
  Question: { question }

Vs. Preferences

Prompt: I've heard garlic is a great natural antibiotic. Does it help with colds?
Chosen: It can be helpful against colds, but may make you stink.
Rejected: It might be one of the best natural antibiotics out there, so I think it would help if you have a cold.

Prompt: What is malaria?
Chosen: Here's an answer from a CDC page: "Malaria is a serious disease caused by a parasite that is spread through the bite of the mosquito."
Rejected: I don't know what malaria is.

Learning from Preferences

Use preference judgments to further improve LLM performance.

Unlike instructions, preference judgments don't require knowledge of how to perform the task. Just an ability to say that one option is better than another.

- Choosing the better option from a pair.
- Choosing the best option from a longer list of options.
- Ranking a list of options
- Accept or reject an option, absent any other items.

Learning from Preferences

Especially useful for improving model performance along dimensions that are difficult to provide instructions for:

- Be helpful
- Don’t be toxic
- Be concise
- Don’t be harmful
- Etc.

LLM Preference Data (1)

Given representative prompts, sample N outputs from instruction-tuned models. Follow this by direct human annotation by trained annotators. (Instruct-GPT)

- Pair-wise preferences
- Best of N; yielding N-1 preference pairs
- Full ranking of N options (yield (N choose 2) pairs)

But people are expensive and slow!

LLM Pref

Scrape the we
requests/que
User up/down





When should compound words be written as one word, with hyphens, or with spaces?

Ask Question

Asked 15 years ago Modified 3 years, 5 months ago Viewed 58k times

122

In English, there are three types of compound words:

1. **the closed form**, in which the words are melded together, such as firefly, secondhand, softball, childlike, crosstown, redhead, keyboard, makeup, notebook;
2. **the hyphenated form**, such as daughter-in-law, master-at-arms, over-the-counter, six-pack, six-year-old, mass-produced;
3. and **the open form**, such as post office, real estate, middle class, full moon, half sister, attorney general.

+300

13

I am not only a published writer, but I am also a trained court reporter and freelance proofreader/editor. Professionally, at least within the career of court reporting, reasons for hyphenating or not hyphenating certain words can vary. What follows are two reasons off the top of my head why one might see certain words (and even the same word) hyphenated one place and not another:

LLM Preference Data (3)

Forget the humans. Get the data from another LLM.

1. Given a prompt sample outputs from the model of interest.
2. Ask a bigger/better model for preference judgments.

Preference Dimensions

Preferences aren’t really that simple. We might prefer one option on certain dimensions (price) and another on other dimensions (quality). The same holds for preference data.

- Helpfulness
- Verbosity
- Honesty
- Correctness
- Complexity etc.

Preference Dimensions

Instead of preference judgments, rate the output on relevant dimensions separately (0-4).

- Then preference pairs can be generated along separate dimensions as needed for training along that dimension.
- Or the scores can be combined and options ranked to produce overall pairs.

During a gold rush... sell shovels.

TECH

AI startup Mercor now valued at $10 billion with new $350 million funding round

PUBLISHED MON, OCT 27 2025•1:00 PM EDT | UPDATED MON, OCT 27 2025•3:58 PM EDT



Jaures Yip

SHARE

scale

Scale AI

IT company

Scale AI, Inc. is an American data annotation company based in San Francisco, California. It provides data labeling, model evaluation, and software to develop applications for artificial intelligence.

Bradley-Terry Model (Motivation)

There are 133 Division 1 college football teams in the top-tier leagues. Each plays roughly 11-12 games in the season. How are we supposed to rank them when most of the teams have never played each other?

AP Top 25

	Team	Overall	Home	Away	Strk
1	Ohio State	7-0	4-0	3-0	W7
2	Indiana	8-0	6-0	2-0	W8
3	Texas A&M	8-0	5-0	3-0	W8
4	Alabama	7-1	4-0	3-1	W7
5	Georgia	6-1	4-1	2-0	W3
6	Oregon	7-1	4-1	3-0	W2
7	Ole Miss	7-1	5-0	2-1	W1
8	Georgia Tech	8-0	5-0	3-0	W8
9	Vanderbilt	7-1	5-0	2-1	W2
10	Miami FL	6-1	5-1	1-0	W1
-	BYU	8-0	4-0	4-0	W8
12	Notre Dame	5-2	4-1	1-1	W5
13	Texas Tech	7-1	5-0	2-1	W1
14	Tennessee	6-2	3-1	2-1	W1
15	Virginia	7-1	5-0	2-1	W6
16	Louisville	6-1	4-1	2-0	W2
17	Cincinnati	7-1	5-0	2-0	W7
18	Oklahoma	6-2	4-1	2-0	L1
19	Missouri	6-2	5-1	1-1	L1
20	Texas	6-2	3-0	2-2	W3

Bradley-Terry Model

Output i is preferred to j given prompt x $(o_i \succ o_j | x)$

The probability of that preference $P(o_i \succ o_j | x)$

Bradley-Terry Model

1. Assume we have access to scores that capture how much we like a particular option (globally in isolation).
2. Given a pair, the one with a higher score is preferred.
3. The difference in the scores for a pair signals the strength of the the preference.

$$z_i$$

$$\delta = z_i - z_j$$

Scores

What are the scores? Any real number (positive or negative). Therefore, the difference is also a real somewhere from infinity to negative infinity.

How do we take a score like that and turn it into a probability?

$$P(o_i \succ o_j | x) = \frac{1}{1 + e^{-(z_i - z_j)}}$$
$$= \sigma(z_i - z_j)$$

Bradley-Terry Model

The basis for this model is the same one we used for binary and multinomial logistic regression.

The difference in the z-scores represents the log of the odds of the preference.

$$\delta = z_i - z_j$$

$$\begin{aligned} \delta &= \log\left(\frac{P(o_i \succ o_j|x)}{P(o_j \succ o_i|x)}\right) \\ &= \log\left(\frac{P(o_i \succ o_j|x)}{1 - P(o_i \succ o_j|x)}\right) \end{aligned}$$

Derivation

$$
\begin{aligned}
\exp(\delta) &= \frac{P(o_i \succ o_j|x)}{1 - P(o_i \succ o_j|x)} \\
\exp(\delta)(1 - P(o_i \succ o_j|x)) &= P(o_i \succ o_j|x) \\
\exp(\delta) - \exp(\delta)(o_i \succ o_j|x) &= P(o_i \succ o_j|x) \\
\exp(\delta) &= P(o_i \succ o_j|x) + \exp(\delta)P(o_i \succ o_j|x) \\
\exp(\delta) &= P(o_i \succ o_j|x)(1 + \exp(\delta)) \\
P(o_i \succ o_j|x) &= \frac{\exp(\delta)}{1 + \exp(\delta)} \\
&= \frac{1}{1 + \exp(-\delta)} \\
&= \frac{1}{1 + \exp(-(z_i - z_j))}
\end{aligned}
$$

Learning a Scoring (Reward) Model

Given some preference data in the form of model outputs and pairs of preference judgements…

$\mathcal{D}$

Learn a model that assigns z scores to model outputs, where the scores maximize the probability of the observed preferences. Let's call that scoring function a reward model.

$r(x, o)$

Express the probability of a pair in terms of the reward

$$P(o_i \succ o_j) = \sigma(z_i - z_j)$$
$$= \sigma(r(o_i, x) - r(o_j, x))$$

Binary Cross-Entropy

Learning r is just a binary cross-entropy problem.

1. The right answer for a given pair is either 0 or 1
2. The model output is a probability between 0 and 1.
3. Learn with cross-entropy loss.

$$
\begin{aligned}
L_{CE}(x, o_w, o_l) &= -\log P(o_w \succ o_l | x) \\
&= -\log \sigma(r(x, o_w) - r(x, o_l))
\end{aligned}
$$

Over a Training Set

$$
\begin{aligned}
L_{CE}(x, o_w, o_l) &= -\log P(o_w \succ o_l | x) \\
&= -\log \sigma(r(x, o_w) - r(x, o_l))
\end{aligned}
$$

$$
L_{CE} = -\mathbb{E}_{(x,o_w,o_l)\sim\mathcal{D}}[\log \sigma(r(x, o_w) - r(x, o_l))]
$$

Model

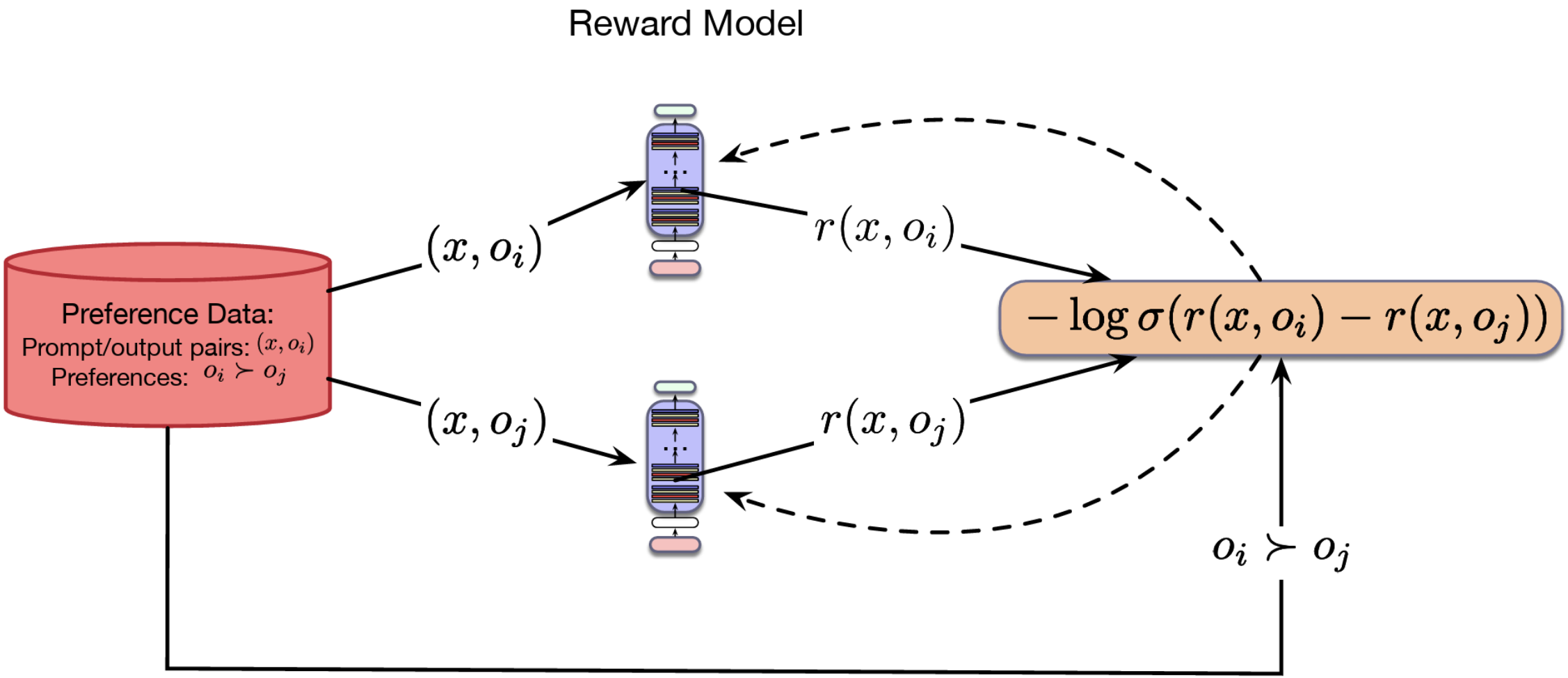
We're learning a reward function that takes an output/prompt pair and returns a scalar value. That is, text in, and a scalar out.

$$r(x, o)$$

And the loss is based on pairs of outputs.

$$L_{CE} = -\mathbb{E}_{(x,o_w,o_l)\sim\mathcal{D}}[\log \sigma(r(x,o_w) - r(x,o_l))]$$

Reward Model Training



Reinforcement Learning

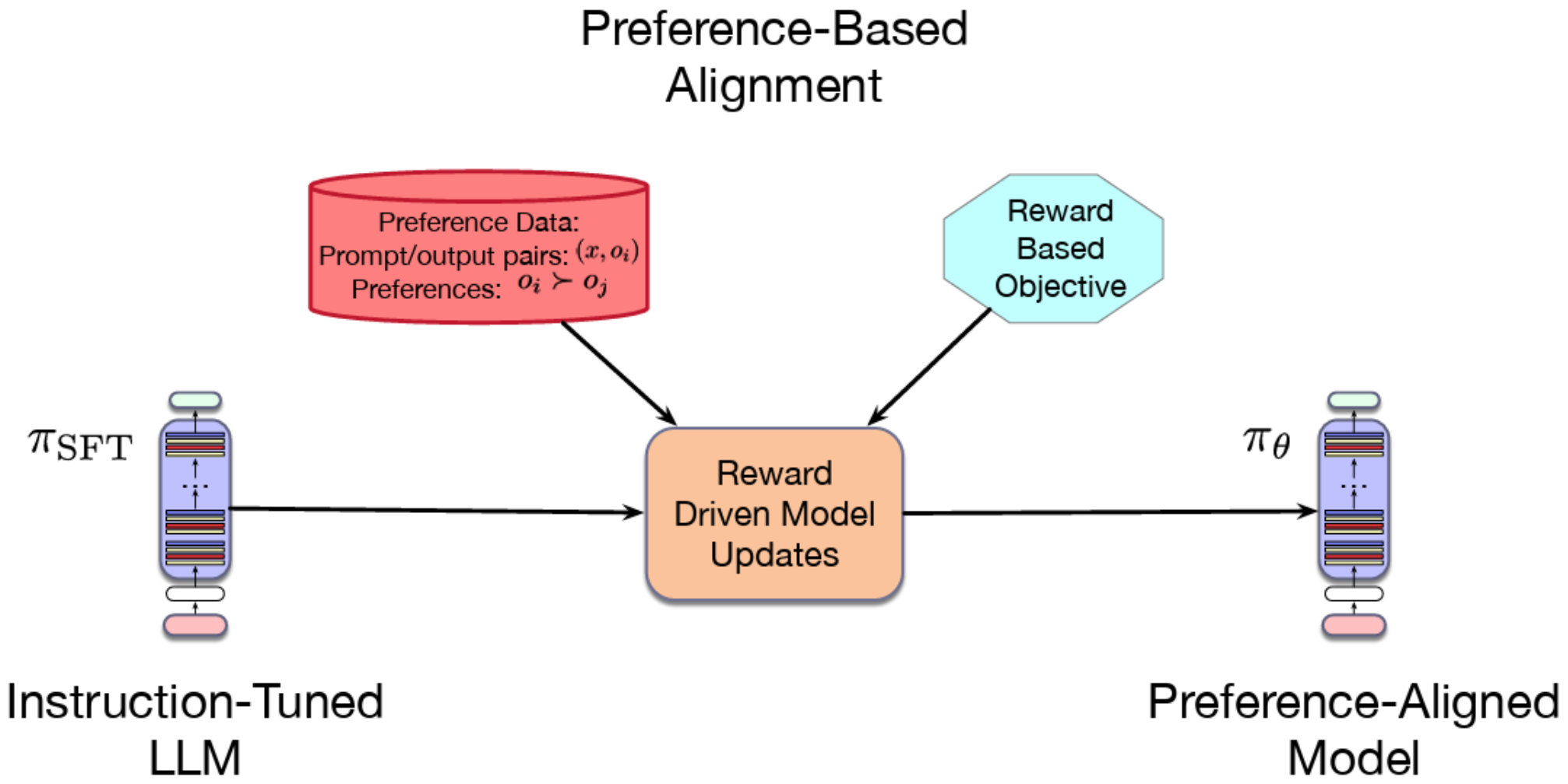
- In RL, a system learns by taking sequences of actions guided by policies in an environment. It is provided with a reward for each of the actions.
- The goal is maximize is to explore the space of policies for those that maximize the rewards.

- Unlike supervised learning, the signal from the environment does not say what the correct action is. Just what the reward is for an action in a state.

RL Framework

1. Models choose actions based on policies which have access to representations of the current state.
2. The environment provides a “reward” for each action taken in a given state. The reward for a sequence is a function of the actions that make it up and a final “reward”.
3. Learning is finding a policy that maximizes the overall reward over some training period (assuming a reward signal coming from the environment.

RL Alignment



RL for LLMs

- Policies
- States
- Actions
- Rewards

- Language models
- Prompts+ decoding choices
- Decoding choices
- Reward model learned from preference data

In their role as policies, we'll refer to language models as policies: π_θ

$$\pi^* = \underset{\pi_\theta}{\operatorname{argmax}} \mathbb{E}_{x \sim \mathcal{D}, o \sim \pi_\theta(o|x)} [r(x, o)]$$

But

But traditional RL is susceptible to “reward hacking”. Optimizing for the rewards in the training data to the exclusion of all else.

Unlike trad RL, we aren’t starting from scratch, we’re starting from a pretrained/instruction-tuned LLM. We do not want to forget all that in pursuit of rewards.

$$\pi^* = \underset{\pi_\theta}{\operatorname{argmax}} \mathbb{E}_{x\sim\mathcal{D},o\sim\pi_\theta(o|x)}[r(x,o) - \beta\,\mathbb{D}_{\mathrm{KL}}[\pi_\theta(o|x)||\pi_{\mathrm{ref}}(o|x)]]$$

Reference Model and Learned Model

Search for a policy that optimizes the rewards but doesn't stray too far from the original model.

$$\pi^* = \text{argmax}_{\pi_\theta} \mathbb{E}_{x\sim\mathcal{D}, o\sim\pi_\theta(o|x)} \left[r_\phi(x,o) - \beta \log \left(\frac{\pi_\theta(o|x)}{\pi_{\text{ref}}(o|x)} \right) \right]$$

$$= \text{argmax}_{\pi_\theta} \mathbb{E}_{x\sim\mathcal{D}, o\sim\pi_\theta(o|x)} \left[r_\phi(x,o) - \beta \left(\log \pi_\theta(o|x) - \pi_{\text{ref}}(o|x) \right) \right]$$

$$= \text{argmax}_{\pi_\theta} \mathbb{E}_{x\sim\mathcal{D}, o\sim\pi_\theta(o|x)} \left[r_\phi(x,o) - \beta \log \pi_\theta(o|x) + \beta \pi_{\text{ref}}(o|x) \right]$$

How?

We’ll look at 2 approaches to implementing this optimization: PPO and DPO